

Jiarui Hai

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RESEARCH INTEREST

My research focuses on AI for audio and speech signal processing, with a particular emphasis on generative models and the integration of natural language prompting. Specifically, I am interested in text-to-audio generation, text-to-speech synthesis, text-guided source separation, and audio understanding tasks such as general audio question answering and text-guided sound event detection.

In addition, my background and passion for music production inspire a complementary interest in music technology, including music generation, transcription, source separation and restoration, and singing voice synthesis.

EDUCATION

- **Johns Hopkins University** Baltimore, MD, USA
Ph.D Student in Electrical and Computer Engineering
Advisor: Prof. Mounya Elhilali Aug 2022 - Expected 2026
- **Tsinghua University** Beijing, China
M.E. in Civil Engineering
Member in Big Data Program Aug 2020 - Jun 2022
- **Tsinghua University** Beijing, China
B.E. in Civil Engineering
B.S. in Business Analytics Aug 2016 - Jun 2020

EXPERIENCE

- **Adobe CAVA | Speech AI Lab** San Francisco, CA, USA
Research Intern | Supervisors: Ke Chen, Rithesh Kumar, Zeyu Jin
Research Topic: Speech Synthesis May 2025 - Present
- **Johns Hopkins University | LCAP** Baltimore, MD, USA
Research Assistant | Supervisor: Mounya Elhilali
Research Topic: Audio Generation and Understanding Aug 2022 - Present
- **Tencent AI Lab | Speech Group** Bellevue, WA, USA
Research Intern | Supervisors: Yong Xu, Hao Zhang, Dong Yu
Research Topic: Audio Generation May 2024 - Aug 2024
- **Kuaishou | AI Platform** Beijing, China
Algorithm Engineer Intern | Supervisors: Rui Lu, Zhiyao Duan
Research Topic: Singing Voice Synthesis Jun 2020 - Jan 2021

SELECTED PUBLICATIONS

FULL LIST AVAILABLE ON GOOGLE SCHOLAR

- [12] **Jiarui Hai**, Helin Wang, Weizhe Guo, Mounya Elhilali. "FlexSED: Towards Open-Vocabulary Sound Event Detection". WASPAA 2025 (Oral).
- [11] **Jiarui Hai**, Mounya Elhilali. "SynSonic: Augmenting Sound Event Detection through Text-to-Audio Diffusion ControlNet and Effective Sample Filtering". WASPAA 2025.
- [10] Yongyi Zang, **Jiarui Hai**, Wanying Ge, Helin Wang, Zheqi Dai, Yuki Mitsufuji, Qiuqiang Kong, Mark Plumbley. "The Inaugural Music Source Restoration Challenge". ICASSP 2026 Grand Challenge.
- [9] Helin Wang*, **Jiarui Hai***, Dading Chong, Karan Thakkar, Tiantian Feng, Dongchao Yang, Junhyeok Lee, Laureano Moro Velazquez, Jesus Villalba, Zengyi Qin, Shrikanth Narayanan, Mounya Elhiali, Najim Dehak. "CapSpeech: Enabling Downstream Applications in Style-Captioned Text-to-Speech". Preprint.
- [8] Helin Wang, **Jiarui Hai**, Dongchao Yang, Chen Chen, Kai Li, Junyi Peng, Thomas Thebaud, Laureano Moro-Velazquez, Jesus Villalba, Najim Dehak. "SoloSpeech: Enhancing Intelligibility and Quality in Target Speech Extraction through a Cascaded Generative Pipeline". Preprint.
- [7] **Jiarui Hai**, Yong Xu, Hao Zhang, Chenxing Li, Helin Wang, Mounya Elhilali, Dong Yu. "EzAudio: Enhancing Text-to-Audio Generation with Efficient Diffusion Transformer". Interspeech 2025 (Oral).
- [6] Helin Wang*, **Jiarui Hai***, Yen-Ju Lu, Karan Thakkar, Mounya Elhilali, Najim Dehak. "SoloAudio: Target Sound Extraction with Language-oriented Audio Diffusion Transformer". ICASSP 2025.
- [5] **Jiarui Hai***, Karan Thakkar*, Helin Wang, Zengyi Qin, Mounya Elhilali. "DreamVoice: Text-Guided Voice Conversion". Interspeech 2024.
- [4] **Jiarui Hai***, Helin Wang*, Dongchao Yang, Karan Thakkar, Najim Dehak, Mounya Elhilali. "DPM-TSE: A Diffusion Probabilistic Model for Target Sound Extraction". ICASSP 2024.

- [3] **Jiarui Hai***, Yu-jeh Liu*, Mounya Elhilali. "Boosting Modality Representation with Pre-trained Models and Multi-task Training for Multimodal Sentiment Analysis". ASRU 2023.
- [2] **Jiarui Hai**, Mounya Elhilali. "Diff-Pitcher: Diffusion-based Singing Voice Pitch Correction". WASPAA 2023 (**Oral**).
- [1] Rui Lu, Baigong Zheng, **Jiarui Hai**, Fei Tao, Zhiyao Duan, Ji Liu. "Progressive Teacher-Student Training Framework for Music Tagging". ICASSP 2022.

RESEARCH ACTIVITIES

- **Peer Reviewer**

- Conference: ICASSP, InterSpeech, WASPAA, ASRU, SLT
- Journal: IJCV
- Workshop: DeLTa @ ICLR 2025, Audio Imagination @ NeurIPS 2024

- **Open-source Contribution**

- **OpenSound** User-friendly live demos for diverse audio and speech models: <https://huggingface.co/OpenSound>
- **MeanFlow-PyTorch** A few-step flow-matching model training pipeline: <https://github.com/haidog-yaqub/MeanFlow>

- **Datasets, Benchmarks, and Challenges**

- **Music Source Restoration Challenge** A benchmark for music source restoration: <https://msrchallenge.com/>
- **CapSpeech** A large-scale speech captioning dataset: <https://wanghelin1997.github.io/CapSpeech-demo/>
- **DreamVoice** A voice timbre dataset: https://haidog-yaqub.github.io/dreamvoice_demo/

MUSIC ACTIVITIES

- **Professional Recognition**

- Featured Producer on a rap single showcased in HIPHOP BANK (TV show, China) | Aug 2021
- Invited Lecture on Music Production, Modern Sky (Chinas leading music company) | Jun 2021
- Top New Producer Award, BeatsHome Hip-hop Production Contest | Apr 2021

- **Creative & Media Activities**

- Independent Artist & Beatmaker: produce singles and beat tapes
- Content Creator: share tutorials on music production and mixing

- **Leadership and Co-founder**

- President, Rap & HipHop Association, Tsinghua University | 2020–2021
- Co-founder, InnerSup Records (Music Producer Team)
- Co-founder, Music Production Team, Rap & HipHop Association, Tsinghua University